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United States Patent
Kind Code
Date of Patent
Inventor(s)

12410902
B1
September 09, 2025
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Vehicle lamp

Abstract

A vehicle lamp comprises a housing including a rear wall and an annular wall located in front of and connected to the rear wall. A chamber between the annular wall and the rear wall includes a front end having an opening. A carrier seat protrudes from an inner surface of the annular wall into the chamber and includes a mounting face facing the chamber. The annular wall includes an assembling opening facing the mounting face. A circuit board disposed on the mounting face includes at least one light emitting element and at least one first mounting hole. An optical element is disposed in the chamber to guide light rays emitted by the at least one light emitting element. A front cover is coupled with the housing and substantially seals the opening. A vertical projection of the at least one first mounting hole is substantially located in the assembling opening.

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Appl. No.: 18/932349

Filed: October 30, 2024

Publication Classification

Int. Cl.: F21S45/48 (20180101); F21S41/20 (20180101); F21S41/39 (20180101); F21S43/20 (20180101); F21S43/37 (20180101); F21V19/00 (20060101); F21Y115/10 (20160101)

U.S. Cl.:

CPC F21S45/48 (20180101); F21S41/2805 (20240501); F21S41/39 (20180101); F21S43/265 (20240501); F21S43/37 (20180101); F21V19/0055 (20130101); F21Y2115/10

Field of Classification Search

CPC: F21S (45/48); F21S (41/2805); F21S (41/39); F21S (41/19); F21S (41/192); F21S (43/265); F21S (43/37); F21S (41/145); F21S (41/147); F21Y (2115/10); F21V (19/0055); F21V (15/01)

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates to a vehicle lamp and, more particularly, to a vehicle lamp which provides an enhanced cooling effect and which is easily assembled.

(2) A conventional vehicle lamp generally includes a cooling device to avoid the light emitting element from being damaged due to overheating. For example, Taiwan Utility Model TWM409195 discloses a vehicle lamp structure mainly including a housing, an LED light source module, a thermally conductive seat, a reflective cover, and a light transmitting cover. The housing is made of better thermally conductive material and includes a receiving portion assembled with a circuit board by fasteners. The thermally conductive seat is made of better thermally conductive material and includes mounting plates securely fixed to the housing by fasteners. The thermally conductive seat is also assembled by fasteners to the LED light source module. The LED light source module is electrically connected to the circuit board, such that the heat generated during operation of the LED light source module can be transferred through the thermally conductive seat to the housing where the heat is dissipated.

(3) However, in the above assembly, an air gap exists in a connection area between the LED light source module and the thermally conductive seat, and another air gap exists in a connection area between the thermally conductive seat and the housing. These two air gaps reduce the thermal transfer effect and, thus, reduce the cooling effect.

(4) Taiwan Utility Model No. TWM496724 discloses a vehicle lamp structure including a housing, a reflective cover, an LED lamp unit, a shielding cover, and a lamp cover. The housing is made of metal and includes a front portion and a rear portion. A platform is disposed on the rear portion and includes a connecting portion connected to by bolts to the LED lamp unit. Then, the front portion is assembled with the rear portion. Next, the reflective cover, the shielding cover, and the lamp cover are assembled in sequence to finish the vehicle lamp.

(5) The housing includes separable front and rear portions, such that when the LED lamp unit is disposed on the platform of the front portion, the platform is not covered to permit use of tools for assembly. Furthermore, after assembly, the front portion is assembled to provide a larger area for the housing for cooling purposes. However, an air gap exists in the connection area between the LED lamp unit and the connecting portion of the platform, and another air gap exists in the connection area between the front portion and the rear portion, leading to a poor cooling effect of the housing.

(6) Taiwan Patent Publication No. 201702519 discloses a vehicle lamp including a reflecting cover, a partitioning board, an LED circuit board, a shielding board, and a front fixing seat. The reflecting cover includes a lower cover and an upper cover. The partitioning board may dissipate heat and includes heat dissipating fins on a rear end thereof. In assembly, the LED circuit board is firstly disposed on the partitioning board. Then, the upper cover is assembled with the lower cover, and the partitioning board is located in a central portion of the upper and lower covers. Next, the shielding board and the front fixing seat are assembled.

(7) However, the heat sources on the LED circuit board are merely cooled by the partitioning board with a small area, resulting in a poor cooling effect.

BRIEF SUMMARY OF THE INVENTION

(8) An objective of the present invention is to provide a vehicle lamp which provides an enhanced cooling effect and which is easily assembled.

(9) In a first aspect, a vehicle lamp according to the present invention is configured to be mounted on a vehicle body. A first direction is defined in a front/rear direction of the vehicle body. The vehicle lamp comprises a housing, a circuit board, an optical element, and a front cover. The housing includes a rear wall and an annular wall located in front of the rear wall and connected to the rear wall. A chamber is defined between the annular wall and the rear wall and includes a front end having an opening. The annular wall includes an inner surface. A carrier seat protrudes from the inner surface into the chamber. The carrier seat includes a mounting face facing the chamber. The annular wall includes an assembling opening facing the mounting face. The circuit board includes at least one light emitting element and at least one first mounting hole. The circuit board is disposed on the mounting face. The optical element is disposed in the chamber and is configured to guide light rays emitted by the at least one light emitting element. The front cover is coupled with the housing. The front cover substantially seals the opening. A vertical projection of the at least one first mounting hole is substantially located in the assembling opening.

(10) In an example, the vehicle lamp further comprises at least one assembling element. The mounting face is provided with at least one second mounting hole. The at least one assembling element extends through the at least one first mounting hole of the circuit board and the at least one second mounting hole of the mounting face of the carrier seat to assemble the circuit board with the carrier seat.

(11) In an example, the assembling opening is in the form of a first notch facing forward.

(12) In another example, the assembling opening is the form of a hole.

(13) In an example, the mounting face is planar and is substantially parallel to the inner surface of the annular wall on which the carrier seat is located.

(14) In an example, the mounting face is planar and is substantially at a non-parallel angle to the inner surface of the annular wall on which the carrier seat is located.

(15) In an example, the housing and the carrier seat are integrally formed and are made of metal or

include a material with high thermal conductivity.

(16) In an example, the housing includes a plurality of heat dissipating fins protruding from an outer surface of the housing.

(17) In an example, the optical element includes at least one first sidewall and at least one second sidewall connected to the at least one first sidewall. The at least one first sidewall and the at least one second sidewall of the optical element are bonded to the inner surface of the annular wall by a glue.

(18) In an example, claim the at least one second sidewall is aligned with the assembling opening.

(19) In an example, the first sidewall of the optical element is disposed annularly and faces the inner surface of the annular wall. Furthermore, an extending direction of the first sidewall is at an angle of about 75 degrees to the first direction.

(20) In an example, the at least one second sidewall includes two second sidewalls protruding toward the inner surface of the annular wall. A spacing between the two second sidewalls decreases gradually toward a rear end of the optical element. An extending direction of each of the two second sidewalls is at an angle of about 30 degrees to the first direction.

(21) In an example, the at least one second sidewall includes two second sidewalls protruding toward the inner surface of the annular wall. A spacing between the two second sidewalls decreases gradually toward a rear end of the optical element. An extending direction of each of the two second sidewalls is at an angle to the first direction. The angle is smaller than 90 degrees and not smaller than 30 degrees.

(22) In an example, the glue is annularly and continuously applied to the inner surface of the annular wall of the housing.

(23) In an example, the front cover includes an outer frame and a light-transmitting lens. The light-transmitting lens is disposed in a central portion of the outer frame. The outer frame includes a shielding portion protruding rearward to cover the assembling opening.

(24) In an example, the optical element is a reflecting cover.

(25) By the provision of the assembling opening, when the circuit board is disposed on the mounting face of the carrier seat, rapid assembly may be achieved by extending the external assembling device through the assembling opening, thereby providing easy assembly. Furthermore, the heat sources generated by the light emitting element may be transferred through the circuit board to the carrier seat and then to the housing for cooling purposes. The heat transfer path includes only one air gap, thereby providing a better heat transfer efficiency. Furthermore, the housing according to the present invention is a single piece with a larger cooling area, thereby enhancing the cooling effect. Furthermore, according to the present invention, the glue is annularly and continuously applied to the inner surface of the annular wall of the housing, such that the housing, the optical element, and the front cover are coupled together as an integral member, thereby providing stable assembly.

(26) In another aspect, a vehicle lamp according to the present invention comprises a housing including a chamber, an inner surface, and an assembling opening. A carrier seat is disposed on the inner surface and includes a mounting face aligned with the assembling opening. A circuit board is disposed in the chamber. An optical element is disposed in the chamber. A front cover substantially covers the chamber. A vertical projection of at least a portion of the mounting face is substantially located in the assembling opening.

(27) In a further aspect, a vehicle lamp according to the present invention comprises a housing including a chamber, an inner surface, and an assembling opening. The inner surface includes a mounting face aligned with the assembling opening. A circuit board is disposed in the chamber. An optical element is disposed in the chamber. At least one assembling element and the circuit board are assembled on the mounting face by an external assembling device through the assembling opening.

(28) The present invention will become clearer in light of the following detailed description of illustrative embodiments of this invention described in connection with the drawings.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded, perspective view of a vehicle lamp according to the present invention.
- (2) FIG. 2 is a schematic view illustrating assembly of a housing and a circuit board of the vehicle lamp according to the present invention.
- (3) FIG. 3 is a schematic view illustrating gluing of the housing and an optical element of the vehicle lamp according to the present invention.
- (4) FIG. 4 is a perspective view of the housing and the optical element of the vehicle lamp according to the present invention after gluing.
- (5) FIG. 5 is a perspective view of the vehicle lamp according to the present invention after assembly.
- (6) FIG. 6 is a cross sectional view of the vehicle lamp according to the present invention.
- (7) FIG. 7 is a schematic view illustrating, in another embodiment of the vehicle lamp according to the present invention, that the mounting face of the housing is parallel to the inner surface.
- (8) FIG. 8 is a schematic view illustrating an assembling opening of the housing in the form of a hole in another embodiment of the vehicle lamp according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (9) FIG. 1 is an exploded, perspective view of a vehicle lamp according to the present invention. FIG. 2 is a schematic view illustrating assembly of a housing and a circuit board of the vehicle lamp according to the present invention. FIG. 3 is a schematic view illustrating gluing of the housing and an optical element of the vehicle lamp according to the present invention. FIG. 4 is a perspective view of the housing and the optical element of the vehicle lamp according to the present invention after gluing. FIG. 5 is a perspective view of the vehicle lamp according to the present invention after assembly. FIG. 6 is a cross sectional view of the vehicle lamp according to the present invention. The vehicle lamp **100** according to the present invention is configured to be disposed on a vehicle body (not shown). A first direction D is defined in the front/rear direction of the vehicle body (not shown).
- (10) The vehicle lamp **100** according to the present invention comprises a housing **1**, a circuit board **2**, an optical element **3**, and a front cover **4**. The housing **1** includes a rear wall **11** and an annular wall **12** located in front of the rear wall **11** and connected to the rear wall **11**. A chamber **13** is defined between the annular wall **12** and the rear wall **11** and includes a front end having an opening **131**. The annular wall **12** includes an inner surface **121** and an outer surface **122**. A carrier seat **14** protrudes from the inner surface **121** into the chamber **13**. The housing **1** and the carrier seat **14** are integrally formed and are made of metal or include a material with high thermal conductivity, such as metal, graphite, carbon tube, diamond, aluminum nitride, silicon carbide, etc. The carrier seat **14** includes a mounting face **141** facing the chamber **13**. In this embodiment, the mounting face **141** is planar. Furthermore, to match with the angle arrangement of the light rays to be emitted, the mounting face **141** is at a non-parallel angle to the inner surface **121** of the annular wall **12** on which the carrier seat **14** is located. In another embodiment, as shown in FIG. 7, the mounting face **141** is substantially parallel to the inner surface **121** of the annular wall **12** on which the carrier seat **14** is located.
- (11) The mounting face **141** is provided with at least one second mounting hole **142**. In this embodiment, the mounting face **141** is spaced from the inner surface **121**, and two second mounting holes **142** are provided and extend from the mounting face **141** toward the inner surface **121**. In another embodiment (not illustrated), the housing **1** has a certain thickness per se, and the

second mounting hole **142** may be directly formed in the inner surface **121**. The annular wall **12** includes an assembling opening **15** which faces the mounting face **141**. The assembling opening **15** is located in the vertical direction of the mounting face **141**. The assembling opening **15** may be in the form of a first notch **151** facing forward. In another embodiment, as shown in FIG. **8**, the assembling opening **15** may be in the form of a hole having an annular periphery without any notches. Furthermore, the housing **1** includes a plurality of heat dissipating fins **16** protruding from the outer surface **122** of the annular wall **12** of the housing **1**. Furthermore, a first receiving groove **17** is provided behind the assembling opening **15**, and a second receiving groove **18** is provided between the carrier seat **14** and the inner surface **121**. The first receiving groove **17** and the second receiving groove **18** are located on two opposite sides of the housing **1**, respectively.

(12) The circuit board **2** includes at least one light emitting element **21** and at least one first mounting hole **22**. The vehicle lamp **100** according to the present invention further comprises the at least one assembling element **23**. In this embodiment, the circuit board **2** includes two first mounting holes **22** associated with the two second mounting holes **142**. A vertical projection of each of the first mounting holes **22** and the second mounting holes **142** is substantially located in the assembling opening **15**. The circuit board **2** can be disposed on the mounting face **141**. As shown in FIG. **2**, an external assembling device **6** can directly extend through the assembling opening **15**, and two assembling elements **23** extend through the first mounting holes **22** of the circuit board **2** and the second mounting holes **142** of the mounting face **141** of the carrier seat **14** to assemble the circuit board **2** with the carrier seat **14**. Thus, the vehicle lamp **100** according to the present invention enables rapid and easy assembly.

(13) The optical element **3** is disposed in the chamber **13** and is configured to guide light rays emitted by the light emitting element **21** to the front of the vehicle lamp **100**. The light emitting element **21** may be a light-emitting diode (LED). In this embodiment, the optical element **3** is a reflecting cover and includes a reflecting face **30** (see FIG. **4**) for reflecting light rays. In another embodiment (not illustrated), the optical element **3** may be a lens for refracting or reflecting light. The optical element **3** includes at least one first sidewall **31** and at least one second sidewall **32** connected to the at least one first sidewall **31**. The first sidewall **31** of the optical element **3** is disposed annularly and faces the inner surface **121** of the annular wall **12**. Furthermore, an extending direction of the first sidewall **31** is at an angle **A1** of about 75 degrees to the first direction **D**. In this embodiment, the optical element **3** includes two second sidewalls **32** protruding toward the inner surface **121** of the annular wall **12**. The two second sidewalls **32** face the inner surface **121** of the annular wall **12**, and the positions of the two second sidewalls **32** are approximately aligned with the assembling opening **15**. Furthermore, the spacing between the two second sidewalls **32** decreases gradually toward a rear end of the optical element **3**. In this embodiment, the extending direction of each of the two second sidewalls **32** is at an angle **A2** of about 30 degrees to the first direction **D**, as shown in FIG. **3**. In another embodiment (not illustrated), the angle **A2** between the extending direction of each of the two second sidewalls **32** and the first direction **D** is smaller than 90 degrees and not smaller than 30 degrees. Furthermore, the optical element **3** is provided with a second notch **33** corresponding to the carrier seat **14**. Two portions of the first sidewall **31** on two sides of the second notch **33** are approximately aligned with the second receiving groove **18**.

(14) The front cover **4** is coupled with the housing **1** and substantially seals the opening **131**. The front cover **4** includes an outer frame **41** and a light-transmitting lens **42** disposed in a central portion of the outer frame **41**. The outer frame **41** includes a shielding portion **43** protruding rearward to cover the assembling opening **15**. Furthermore, the outer frame **41** is provided with an annular plate portion **44** which protrudes rearward and which may be inserted between the optical element **3** and the inner surface **121** of the housing **1**. In another embodiment (not illustrated), the shielding portion **43** may be a separate element for covering the assembling opening **15**.

(15) The housing **1**, the optical element **3**, and the front cover **4** are bonded together by a glue **5**.

Specifically, the first and second sidewalls **31** and **32** of the optical element **3** are bonded to the inner surface **121** of the annular wall **12** of the housing **1** by the glue **5**. According to the present invention, with reference to FIGS. **3** and **4**, the glue **5** may be applied by an external glue applying machine **7**. With the optical element **3** disposed in the chamber **13** of the housing **1**, the glue applying machine **7** extends through the opening **131**, such that an end of the glue applying machine **7** may move along the first and second sidewalls **31** and **32** to spray the glue **5**. When the glue applying machine **7** moves to the second sidewalls **32**, the glue applying machine **7** may follow the inclination angle of the second sidewalls **32** to spray the glue **5** without interruption. When the glue applying machine **7** moves to a position behind the assembling opening **15**, the glue **5** may be applied to an area where the first receiving groove **17** locates. When the glue applying machine **7** moves to another position aligned with the second notch **33**, the glue **5** may be applied to an area where the second receiving groove **18** locates. Therefore, according to the present invention, application of the glue **5** is not affected by the assembling opening **15** and the second notch **33**, and the glue **5** is annularly and continuously applied to the inner surface **121** of the annular **12** of the housing **1**, thereby increasing the bonding effect and airtightness between the housing **1**, the optical element **3**, and the front cover **4**.

(16) As shown in FIGS. **5** and **6**, before the glue **5** is dried and hardens, the front cover **4** is mounted to the front end of the housing **1** and substantially seals the opening **131**. The shielding portion **43** may cover the assembling opening **15**. Furthermore, the annular plate portion **44** may be bonded between the inner surface **121** of the housing **1** and the optical element **3** and in the second receiving groove **18** by the glue **5**. Furthermore, the rear end of the shielding portion **43** may be inserted into the first receiving groove **17** and may be bonded by the glue **5**. After the glue **5** is dried and hardens, the housing **1**, the optical element **3**, and the front cover **4** are stably coupled as an integral member.

(17) According to the present invention, by the provision of the assembling opening **15**, when the circuit board **2** is disposed on the mounting face **141** of the carrier seat **14**, rapid and easy assembly may be achieved by extending the external assembling device **6** through the assembling opening **15**. Furthermore, when the light emitting element **21** emits light, the light rays may be reflected by the reflecting face **30** to the front of the vehicle lamp **100** while providing a predetermined lighting pattern. The heat sources of the light emitting element **21** may be transferred through the circuit board **2** to the carrier seat **14** and, thus, dissipated by the housing **1**. Therefore, according to the present invention, since the carrier seat **14** is a part of the housing **1**, when the heat generated by the light emitting element **21** is transferred through the gap between the circuit board **2** and the mounting face **141** to the carrier seat **14**, the heat is dissipated outward through the housing **1**. The heat transfer path includes only one air gap, thereby providing a better heat transfer efficiency. Furthermore, the housing **1** according to the present invention is a single piece with a larger cooling area, thereby enhancing the cooling effect.

(18) In view of the foregoing, the vehicle lamp **100** according to the present invention provides enhanced cooling efficiency and easy and stable assembly.

(19) The foregoing embodiments are examples of the present invention, not limitation to the present invention. All equivalent changes made in accordance with the spirit of the present invention should also fall within the scope of the present invention.

Claims

1. A vehicle lamp, with a first direction defined in a front/rear direction, with the vehicle lamp comprising: a housing including a rear wall and an annular wall located in front of the rear wall and connected to the rear wall, wherein a chamber is defined between the annular wall and the rear wall and includes a front end having an opening, wherein the annular wall includes an inner surface, wherein a carrier seat protrudes from the inner surface into the chamber, wherein the carrier seat

includes a mounting face facing the chamber, and wherein the annular wall includes an assembling opening facing the mounting face; a circuit board including at least one light emitting element and at least one first mounting hole, wherein the circuit board is disposed on the mounting face; an optical element disposed in the chamber and configured to guide light rays emitted by the at least one light emitting element; and a front cover coupled with the housing, wherein the front cover substantially seals the opening, and wherein a vertical projection of the at least one first mounting hole is substantially located in the assembling opening.

2. The vehicle lamp as claimed in claim 1, further comprising at least one assembling element, the at least one assembling element comprising bolts, screws, and/or nail, wherein the mounting face is provided with at least one second mounting hole, wherein the at least one assembling element extends through the at least one first mounting hole of the circuit board and the at least one second mounting hole of the mounting face of the carrier seat to assemble the circuit board with the carrier seat.

3. The vehicle lamp as claimed in claim 1, wherein the assembling opening is a first notch extending in a forward direction along the annular wall.

4. The vehicle lamp as claimed in claim 1, wherein the assembling opening is a form of a hole.

5. The vehicle lamp as claimed in claim 1, wherein the mounting face is planar and is substantially parallel to the inner surface of the annular wall on which the carrier seat is located.

6. The vehicle lamp as claimed in claim 1, wherein the mounting face is planar and is substantially at a non-parallel angle to the inner surface of the annular wall on which the carrier seat is located.

7. The vehicle lamp as claimed in claim 1, wherein the housing and the carrier seat are integrally formed and are made of metal or include a material with high thermal conductivity.

8. The vehicle lamp as claimed in claim 1, wherein the housing includes a plurality of heat dissipating fins protruding from an outer surface of the housing.

9. The vehicle lamp as claimed in claim 1, wherein the optical element includes at least one first sidewall and at least one second sidewall connected to the at least one first sidewall, and wherein the at least one first sidewall and the at least one second sidewall of the optical element are bonded to the inner surface of the annular wall by a glue.

10. The vehicle lamp as claimed in claim 9, wherein the at least one second sidewall is aligned with the assembling opening.

11. The vehicle lamp as claimed in claim 9, wherein the at least one second sidewall includes two second sidewalls protruding toward the inner surface of the annular wall, wherein a spacing between the two second sidewalls decreases gradually toward a rear end of the optical element, and wherein an extending direction of each of the two second sidewalls is at an angle of about 30 degrees to the first direction.

12. The vehicle lamp as claimed in claim 9, wherein the at least one second sidewall includes two second sidewalls protruding toward the inner surface of the annular wall, wherein a spacing between the two second sidewalls decreases gradually toward a rear end of the optical element, wherein an extending direction of each of the two second sidewalls is at an angle to the first direction, and wherein the angle is smaller than 90 degrees and not smaller than 30 degrees.

13. The vehicle lamp as claimed in claim 9, wherein the glue is annularly and continuously applied to the inner surface of the annular wall of the housing.

14. The vehicle lamp as claimed in claim 1, wherein the front cover includes an outer frame and a light-transmitting lens, wherein the light-transmitting lens is disposed in a central portion of the outer frame, and wherein the outer frame includes a shielding portion protruding rearward to cover the assembling opening.

15. The vehicle lamp as claimed in claim 1, wherein the optical element is a reflecting cover.

16. A vehicle lamp comprising: a housing including a chamber, an inner surface, and an assembling opening, wherein a carrier seat is disposed on the inner surface and includes a mounting face aligned with the assembling opening; a circuit board disposed in the chamber; an optical element

disposed in the chamber; and a front cover substantially covers the chamber, wherein a vertical projection of at least a portion of the mounting face is substantially located in the assembling opening.

17. The vehicle lamp as claimed in claim 16, wherein the mounting face is provided with at least one mounting hole, and wherein a vertical projection of the at least one mounting hole of the mounting face is substantially located in the assembling opening.

18. The vehicle lamp as claimed in claim 17, further comprising at least one assembling element, the at least one assembling element comprising bolts, screws, and/or nails, wherein the at least one assembling element extends through the at least one mounting hole of the mounting face to fix the circuit board to the mounting face.

19. The vehicle lamp as claimed in claim 16, wherein the carrier seat protrudes from the inner surface into the chamber, wherein the housing and the carrier seat are integrally formed and are made of metal or include a material with high thermal conductivity.

20. A vehicle lamp comprising: a housing including a chamber, an inner surface, and an assembling opening, wherein the inner surface includes a mounting face aligned with the assembling opening; a circuit board disposed in the chamber; an optical element disposed in the chamber; and at least one assembling element, the at least one assembling element comprising bolts, screws, and/or nails; wherein the at least one assembling element and the circuit board are assembled on the mounting face by an external assembling device through the assembling opening, wherein the external assembling device is a tool for fastening the at least one assembling element.
